

Data Management  
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# Data Warehouse

techniques and tools



SAPIENZA  
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# Data Warehouse

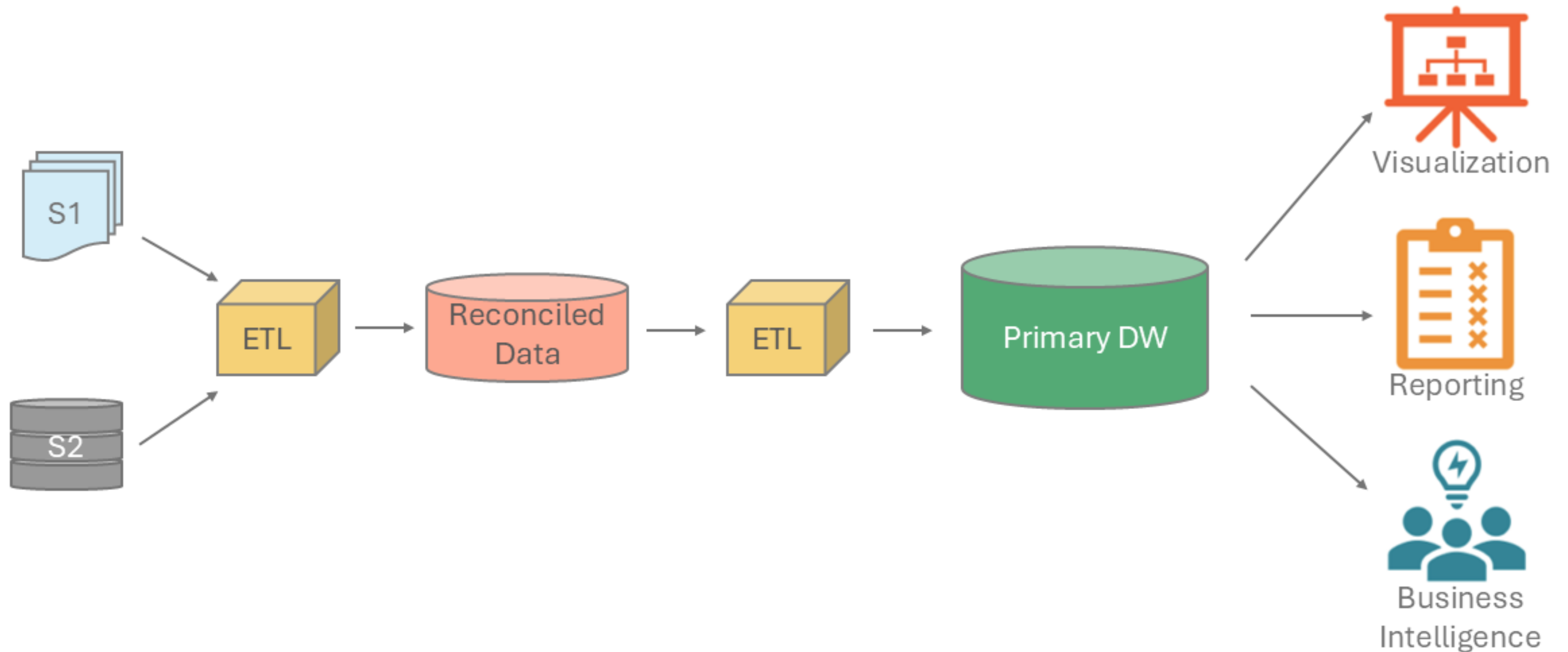
Repository of data to support knowledge workers in strategic decision-making processes

Main pillars:

- Subject-oriented
- Integrated and consistent
- Evolution over time
- Non volatile



# The Data Warehouse Big Picture (3-Layer Architecture)

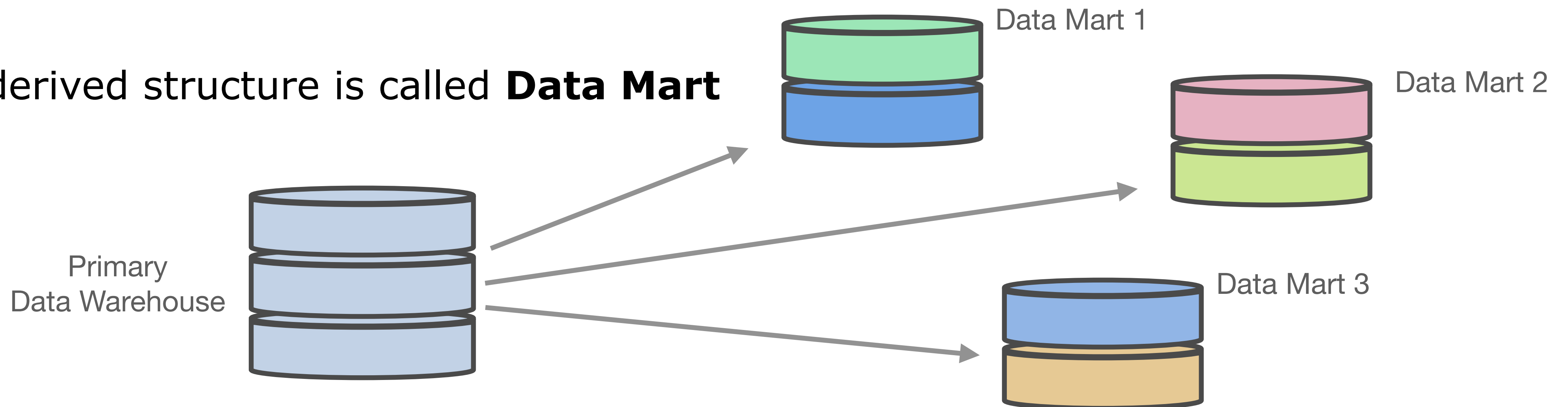


# Data Mart

A Primary Data Warehouse consists of large amount of data involving several aspects of a given business domain: set of subject-oriented DFM/StarSchemas (if ROLAP)

In most cases analyses are made on structures based on specific portions of the data contained in the Primary Data Warehouse

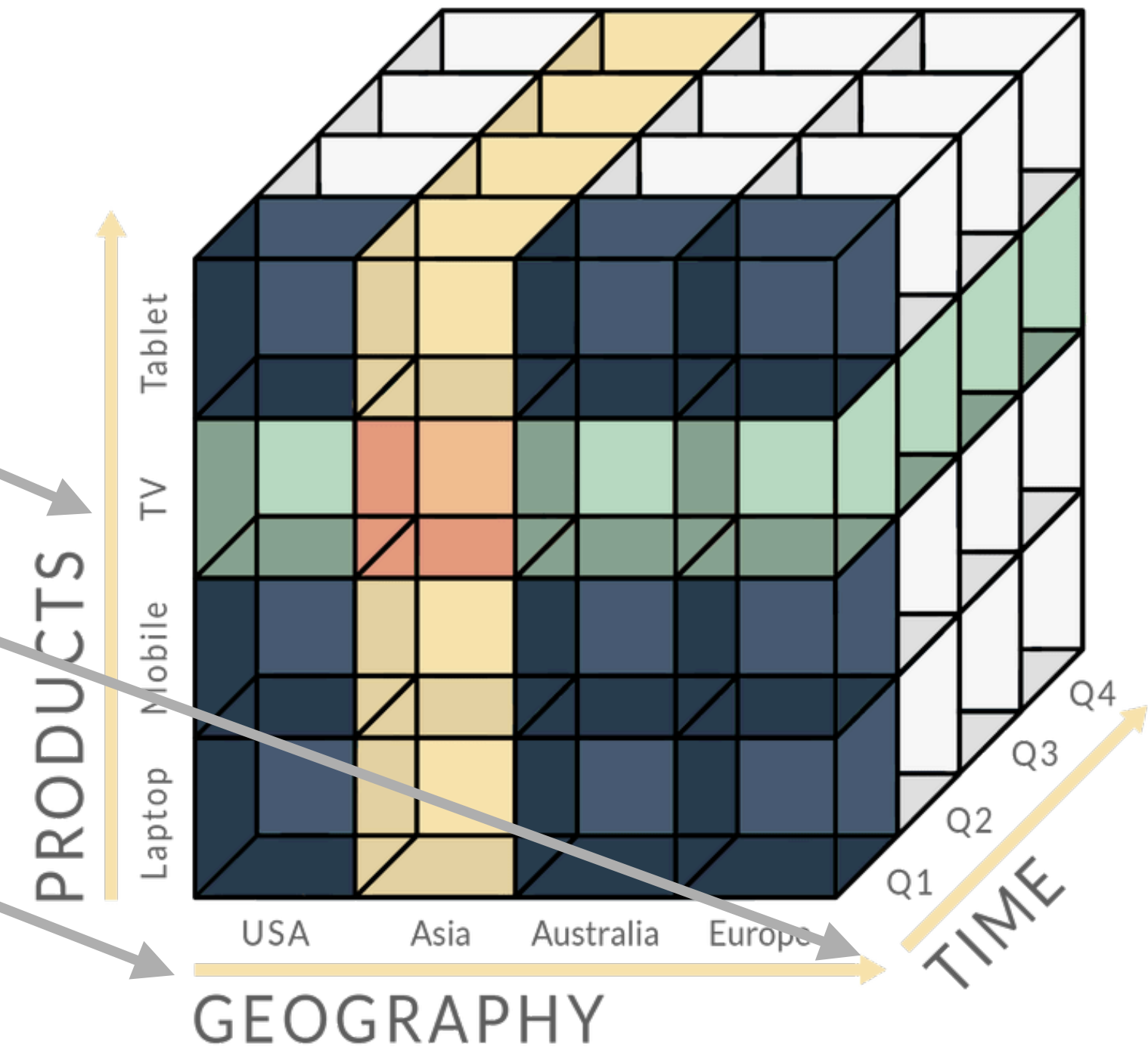
Each derived structure is called **Data Mart**



# Multidimensional model

Data virtually represented as hypercubes (dimensions, facts, measures)

| ID  | Geo    | Price | Time | Qty | Prod   | Disc |
|-----|--------|-------|------|-----|--------|------|
| A10 | USA    | 400   | Q1   | 3   | Mobile | 20%  |
| B13 | Asia   | 600   | Q3   | 7   | Tablet | 0%   |
| D31 | USA    | 900   | Q2   | 2   | TV     | 18%  |
| C7  | Europe | 500   | Q1   | 13  | Mobile | 0%   |
| F9  | Europe | 500   | Q4   | 5   | TV     | 10%  |
| E11 | Asia   | 1000  | Q4   | 4   | Laptop | 12%  |



# Multidimensional model - Aggregation

Suppose we want to know the average base price and average discount applied to sold products for continents and quarters

| Geo    | Time | Prod   | avg(Price) | avg(Disc) |
|--------|------|--------|------------|-----------|
| USA    | Q1   | Mobile | 841\$      | 20%       |
| USA    | Q1   | Tablet | 765\$      | 23%       |
| ...    | ...  | ...    | ...        | ...       |
| Europe | Q3   | Mobile | 698\$      | 13%       |
| Europe | Q4   | TV     | 453\$      | 16%       |
| ...    | ...  | ...    | ...        | ...       |

```
SELECT Geo, Time, Prod, avg(Price), avg(Disc)
FROM T
GROUP BY Geo, Time, Prod
```

- Dimensions
- Measures

# Hierarchies

Axes of hypercubes are called the dimensions and they are used as the coordinates needed to identify portions of the multidimensional data structure containing measures

In some cases dimensions can be organised in **hierarchies**, whose layers allow to express higher or lower levels of granularity in terms of aggregation

Location hierarchy: *Country* → *Region* → *Province* → *City* → *District*

OLAP operations can make use of hierarchies to make analyses based on different levels of detail

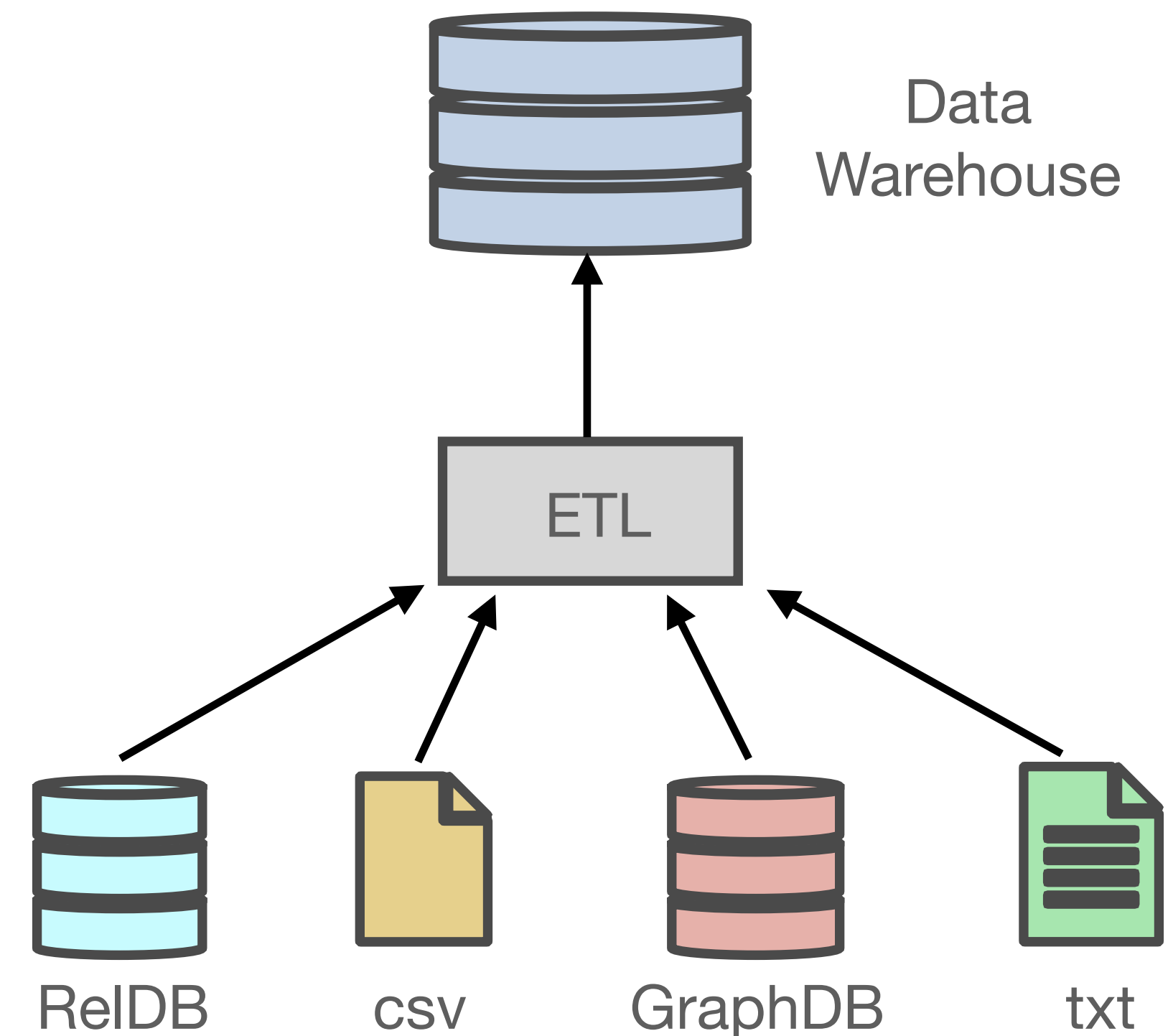
# Integration and ETL (Extract, Transform, Load)

Data sources used to populate a Data Warehouse may be highly heterogeneous (relational DBs, csv, XML, unstructured data, etc.)

Integrating such data is not a trivial task:

- missing data
- duplicate values
- incorrect values
- different formats
- entity resolution
- target schema definition

ETL: plethora of techniques to address such problems



# OLAP session

Process made of subsequent analytical steps where each step is the result of the outcome of preceding ones

An OLAP session is like a navigation path consisting of facts' analyses based on different view points, obtained by means of sequences of queries which are executed on the basis of the OLAP operators used at each step (e.g., by changing aggregation granularity)

Session example:

**S1**: average interest rate of mortgages granted in Italy

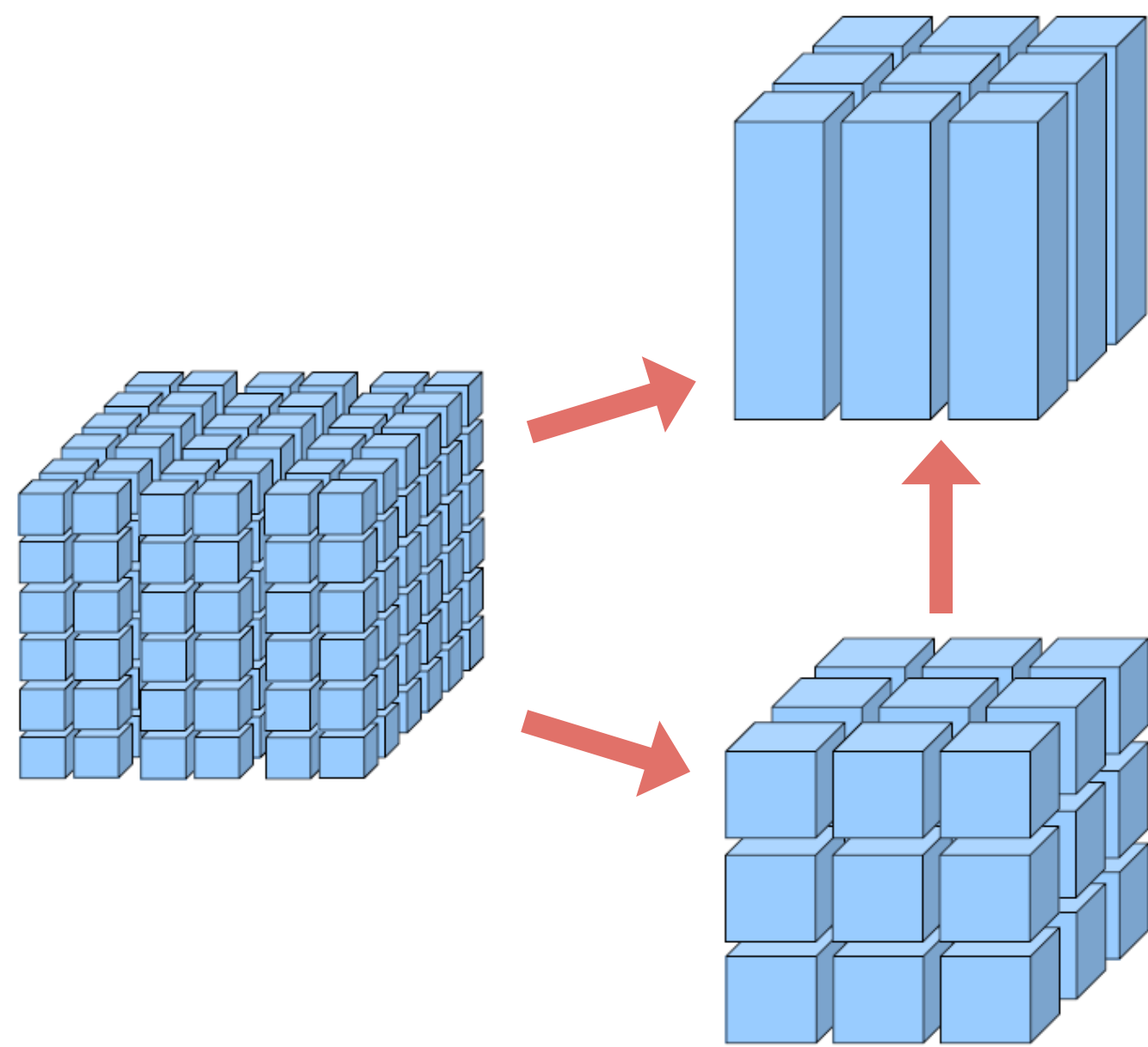
**S2**: average interest rate of mortgages granted in Italy for each Region

**S3**: average interest rate of mortgages granted for each Region, for Education level

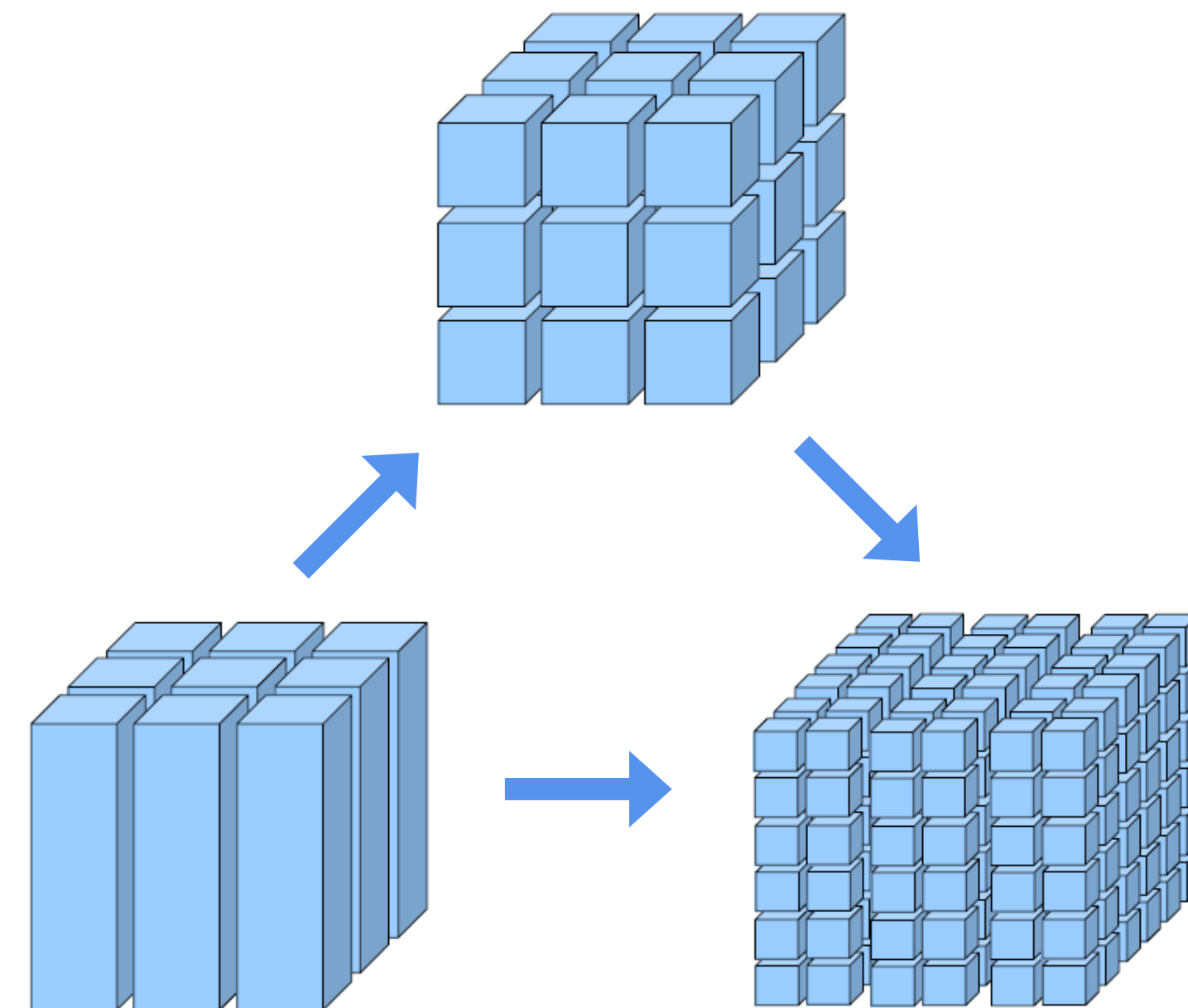
# Roll-up and Drill-down

**Roll-Up:** Aggregates on hierarchies (less detailed cube)

**Drill-Down:** Disaggregates on hierarchies (more detailed cube)



Roll-Up Operations



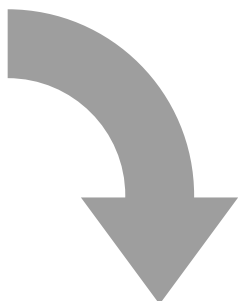
Drill-Down Operations

# Roll-up and Drill-down

**Roll-Up**: Aggregates on hierarchies (less detailed cube)

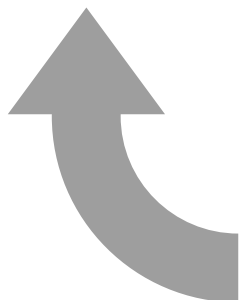
**Drill-Down**: Disaggregates on hierarchies (more detailed cube)

| Time   | North  | Center | South  |
|--------|--------|--------|--------|
| Jan 23 | 96,000 | 83,000 | 9,500  |
| Mar 23 | 58,000 | 81,300 | 53,000 |
| Jul 22 | 31,100 | 27,000 | 76,000 |
| Feb 21 | 23,400 | 31,700 | 43,000 |
| Feb 22 | 64,000 | 57,300 | 61,800 |
| ...    | ...    | ...    | ...    |



**Roll-Up** on Time Hierarchy  
**Drill-Down** on Location Hierarchy

| Time  | FVG   | Veneto | ... | Lazio | ... | Apulia | Sicily |
|-------|-------|--------|-----|-------|-----|--------|--------|
| Q1 23 | 6,000 | 3,500  | ... | 3,500 | ... | 3,100  | 6,200  |
| Q1 22 | 1,200 | 4,500  | ... | 7,300 | ... | 4,500  | 3,800  |
| Q2 22 | 1,700 | 4,200  | ... | 3,200 | ... | 3,700  | 3,100  |
| Q3 22 | 5,200 | 1,600  | ... | 8,200 | ... | 4,100  | 2,600  |
| Q4 22 | 6,400 | 5,700  | ... | 2,300 | ... | 7,800  | 5,300  |
| ...   | ...   | ...    | ... | ...   | ... | ...    | ...    |



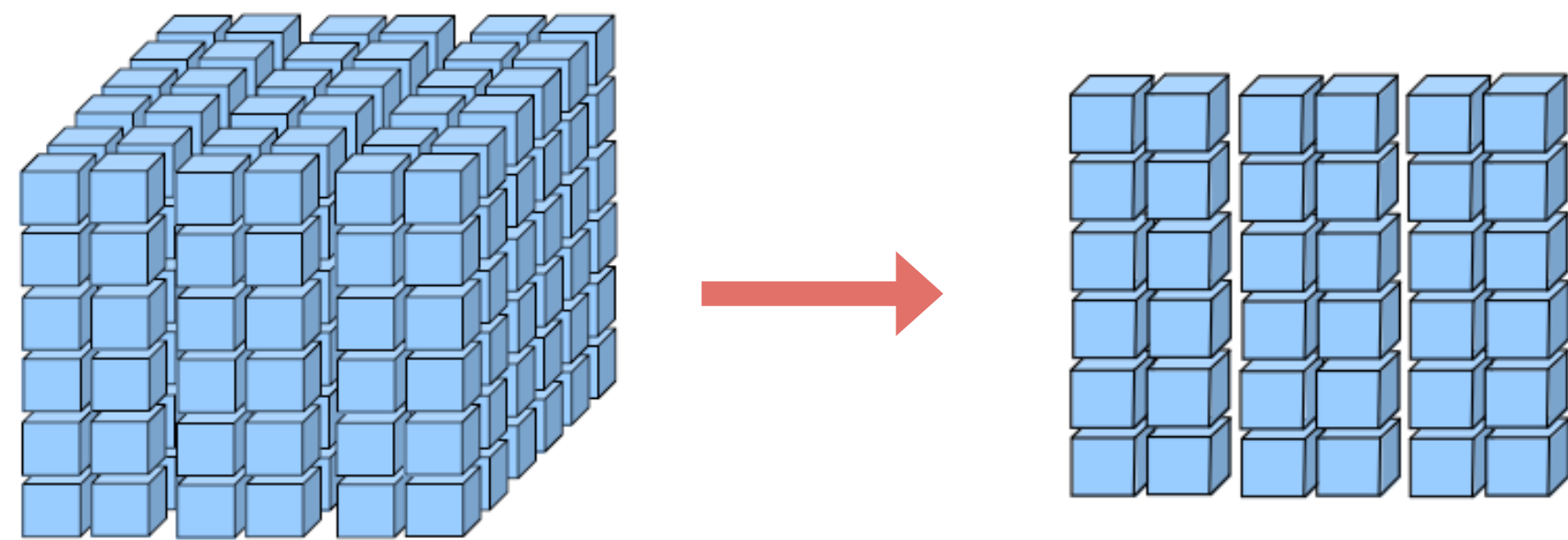
**Roll-Up** on Location Hierarchy  
**Drill-Down** on Time Hierarchy

# Slice and Dice

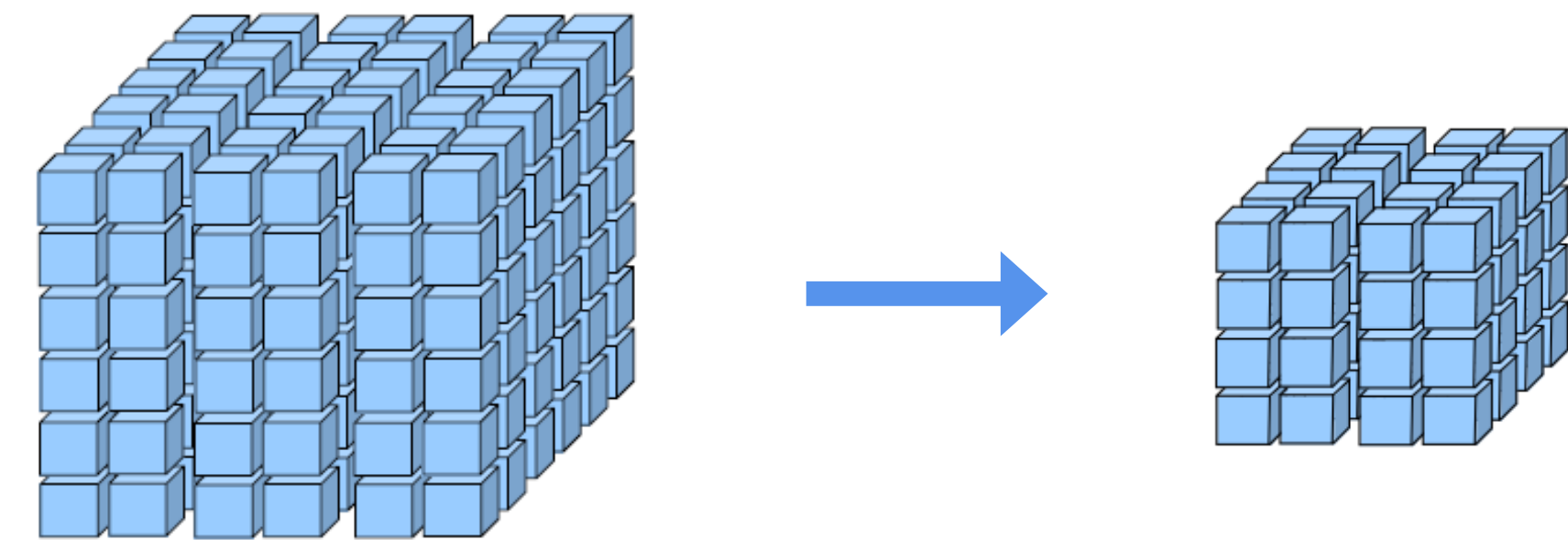
**Slice**: Reduces the number of dimensions by fixing one of them to a specific value

**Dice**: Uses selection criteria over a set of dimensions

Both slice and dice produce sub-cubes



Slice Operation



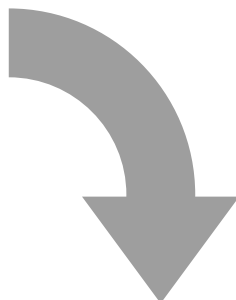
Dice Operation

# Slice and Dice

**Slice:** Reduces the number of dimensions by fixing one of them to a specific value

**Dice:** Uses selection criteria over a set of dimensions

| Quarter | Category   | North  | Center | South  |
|---------|------------|--------|--------|--------|
| Q1      | Vegetables | 96,000 | 83,000 | 9,500  |
|         | Meat       | 58,000 | 81,300 | 53,000 |
| Q2      | Vegetables | 31,100 | 27,000 | 76,000 |
|         | Meat       | 23,400 | 31,700 | 43,000 |
| Q3      | Vegetables | 64,000 | 57,300 | 61,800 |
| ...     | ...        | ...    | ...    | ...    |



**Slice** on Category='Vegetables'

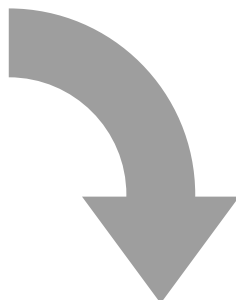
| Quarter | Category   | North  | Center | South  |
|---------|------------|--------|--------|--------|
| Q1      | Vegetables | 96,000 | 83,000 | 9,500  |
| Q2      | Vegetables | 31,100 | 27,000 | 76,000 |
| Q3      | Vegetables | 64,000 | 57,300 | 61,800 |
| ...     | ...        | ...    | ...    | ...    |

# Slice and Dice

**Slice**: Reduces the number of dimensions by fixing one of them to a specific value

**Dice**: Uses selection criteria over a set of dimensions

| Quarte | Cat  | FVG   | Venet | ... | Lazio | ... | Sicily |
|--------|------|-------|-------|-----|-------|-----|--------|
| Q1     | Veg  | 6,000 | 3,500 | ... | 3,500 | ... | 3,100  |
|        | Meat | 1,200 | 4,500 | ... | 7,300 | ... | 4,500  |
| Q2     | Veg  | 1,700 | 4,200 | ... | 3,200 | ... | 3,700  |
|        | Meat | 5,200 | 1,600 | ... | 8,200 | ... | 4,100  |
| Q3     | Veg  | 6,400 | 5,700 | ... | 2,300 | ... | 7,800  |
| ...    | ...  | ...   | ...   | ... | ...   | ... | ...    |



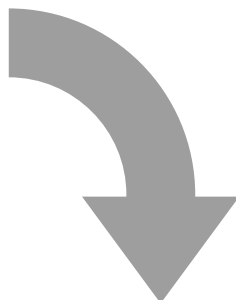
**Dice** on Category='Meat'  $\wedge$  Area='Center'  
 $\wedge$  Rev>1000

| Quarter | Lazio | Umbria | Marche | ... |
|---------|-------|--------|--------|-----|
| Q1      | 7,300 | 4,100  | 2,500  | ... |
| Q2      | 8,200 | 3,700  | 1,800  | ... |
| Q3      | 1,300 | 2,400  | 2,100  | ... |
| ...     | ...   | ...    | ...    | ... |

# Pivoting

Pivoting is a mere visual operation that does not change anything but the structure of the visualisation

| Quarter | Category   | North  | Center | South  |
|---------|------------|--------|--------|--------|
| Q1      | Vegetables | 96,000 | 83,000 | 9,500  |
|         | Meat       | 58,000 | 81,300 | 53,000 |
| Q2      | Vegetables | 31,100 | 27,000 | 76,000 |
|         | Meat       | 23,400 | 31,700 | 43,000 |
| Q3      | Vegetables | 64,000 | 57,300 | 61,800 |
| ...     | ...        | ...    | ...    | ...    |



**Pivoting** on Category

| Quarter | North  |        | Center |        |
|---------|--------|--------|--------|--------|
|         | Vegs   | Meat   | Vegs   | Meat   |
| Q1      | 96,000 | 58,000 | 83,000 | 9,500  |
| Q2      | 31,100 | 23,400 | 27,000 | 31,700 |
| Q3      | 64,000 | 45,000 | 57,300 | 51,000 |
| ...     | ...    | ...    | ...    | ...    |

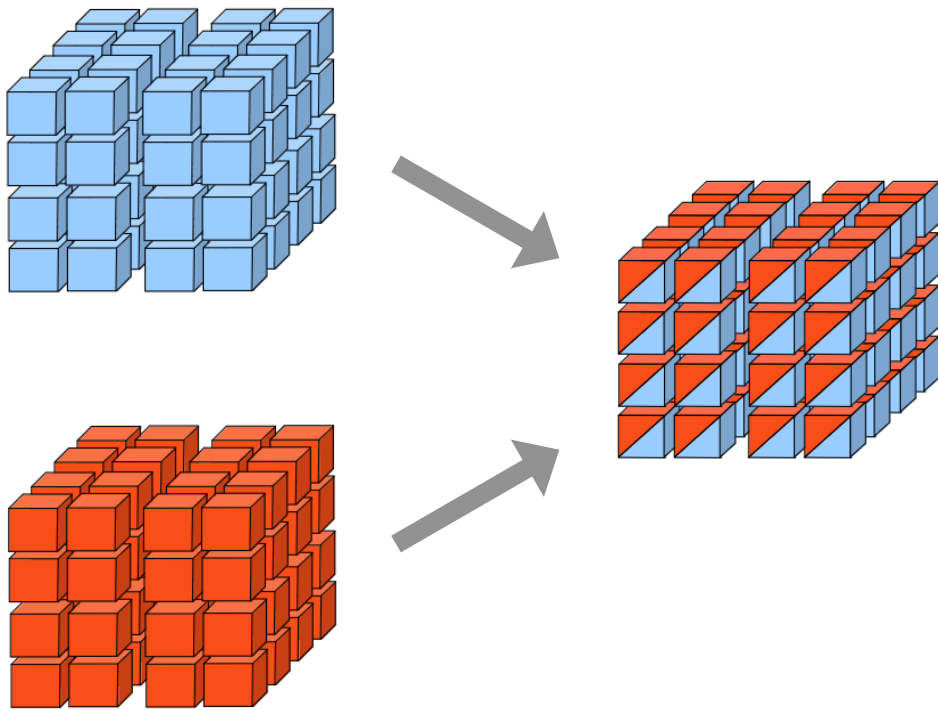
# Drill-across

It is the operation of 'merging' different cubes with the same aggregation level

| Time   | North  | Center | South  |
|--------|--------|--------|--------|
| Jan 21 | 96,000 | 83,000 | 9,500  |
| Mar 23 | 58,000 | 81,300 | 53,000 |
| Jul 22 | 31,100 | 27,000 | 76,000 |
| Feb 21 | 23,400 | 31,700 | 43,000 |
| Feb 22 | 64,000 | 57,300 | 61,800 |
| ...    | ...    | ...    | ...    |

| Time   | North | Center | South |
|--------|-------|--------|-------|
| Jan 21 | 0.2   | 0.3    | 0.13  |
| Mar 23 | 0.21  | 0.25   | 0.4   |
| Jul 22 | 0.15  | 0.1    | 0.18  |
| Feb 21 | 0.3   | 0.2    | 0.5   |
| Feb 22 | 0.22  | 0      | 0.35  |
| ...    | ...   | ...    | ...   |

| Time   | North |        | Center |        | South |        |
|--------|-------|--------|--------|--------|-------|--------|
|        | Disc  | Rev    | Disc   | Rev    | Disc  | Rev    |
| Jan 21 | 0.2   | 83,000 | 0.3    | 83,000 | 0.13  | 9,500  |
| Mar 23 | 0.21  | 81,300 | 0.25   | 81,300 | 0.4   | 53,000 |
| Jul 22 | 0.15  | 27,000 | 0.1    | 27,000 | 0.18  | 76,000 |
| ...    | ...   | ...    | ...    | ...    | ...   | ...    |



# Data Warehousing in practice - Tools

Several tools exist to apply DW techniques

In some cases such tools address specific layers of the DW stacks (ETL, visualisations, BI tools, etc)

Non-exhaustive list of Data Warehousing tools:

- *Amazon Redshift*
- *Google Big Query*
- *Microsoft Power BI*
- *Pentaho*
- *Snowflake*
- *Tableau Public*
- *QlikView*



# Tableau Public

Free (and limited) Data Warehousing platform

Focus on data visualisations for business intelligence

Allows to easily manipulate and understand the main components of the DFM model

Includes interactive dashboards and reports



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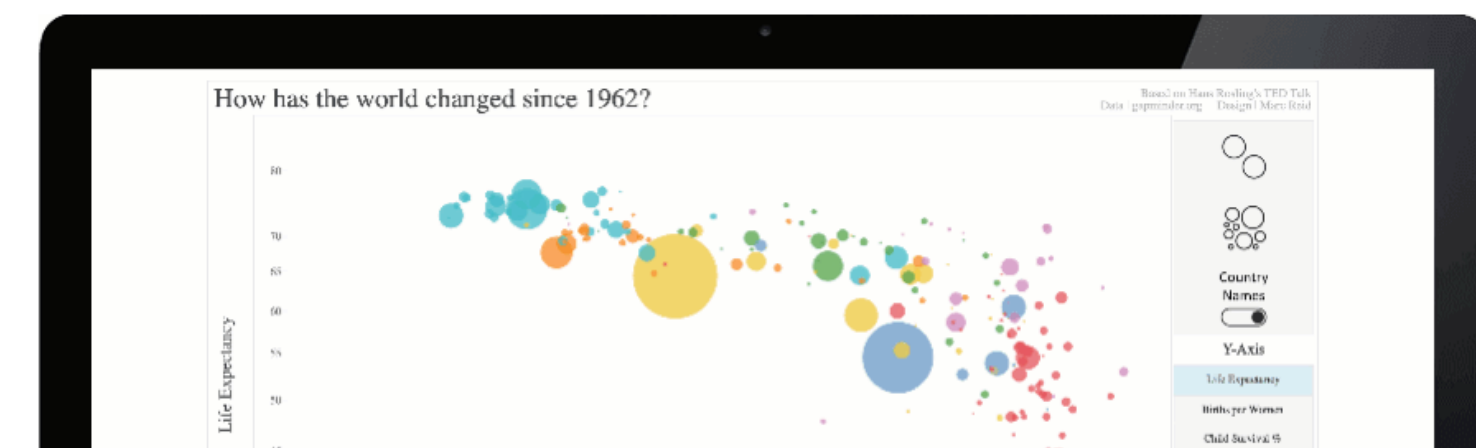
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# Tableau Workspace

Dimensions

Measures

BI Visualisations

| Geographic region                             | Sector of institution         | Applicants total | Bachelor's degrees aw.. | Master's degrees aw.. | Doctor's degrese - r.. | Part-time enrollment | Number of first-time u.. |
|-----------------------------------------------|-------------------------------|------------------|-------------------------|-----------------------|------------------------|----------------------|--------------------------|
| Far West AK CA HI NV OR WA                    | Private not-for-profit, 4-y.. | 370,836          | 46,560                  | 36,046                | 2,876                  | 73,647               | 1,589                    |
|                                               | Public, 4-year or above       | 1,047,737        | 183,295                 | 42,147                | 6,217                  | 175,172              | 2,584                    |
| Great Lakes IL IN MI OH WI                    | Private not-for-profit, 4-y.. | 541,518          | 81,229                  | 41,203                | 2,164                  | 139,682              | 1,739                    |
|                                               | Public, 4-year or above       | 707,526          | 180,256                 | 58,760                | 7,372                  | 263,013              | 6,522                    |
| Mid East DE DC MD NJ NY PA                    | Private not-for-profit, 4-y.. | 1,163,397        | 131,729                 | 95,393                | 6,204                  | 248,300              | 5,558                    |
|                                               | Public, 4-year or above       | 876,695          | 164,776                 | 48,681                | 4,540                  | 257,772              | 2,469                    |
| New England CT ME MA NH RI VT                 | Private not-for-profit, 4-y.. | 657,402          | 59,267                  | 40,517                | 3,263                  | 82,493               | 2,101                    |
|                                               | Public, 4-year or above       | 236,176          | 46,829                  | 12,120                | 1,027                  | 74,421               | 699                      |
| Plains IA KS MN MO NE ND SD                   | Private not-for-profit, 4-y.. | 213,742          | 48,223                  | 22,935                | 693                    | 100,900              | 856                      |
|                                               | Public, 4-year or above       | 263,387          | 87,068                  | 24,054                | 3,531                  | 153,167              | 2,353                    |
| Rocky Mountains CO ID MT UT WY                | Private not-for-profit, 4-y.. | 54,582           | 21,137                  | 8,839                 | 228                    | 30,875               | 182                      |
|                                               | Public, 4-year or above       | 159,457          | 51,212                  | 13,876                | 1,686                  | 133,740              | 1,236                    |
| Southeast AL AR FL GA KY LA MS NC SC TN VA WV | Private not-for-profit, 4-y.. | 751,845          | 89,295                  | 41,490                | 2,888                  | 150,639              | 1,523                    |
|                                               | Public, 4-year or above       | 1,137,213        | 291,417                 | 91,204                | 10,136                 | 452,011              | 5,891                    |
| Southwest AZ NM OK TX                         | Private not-for-profit, 4-y.. | 178,949          | 24,340                  | 10,959                | 535                    | 34,100               | 970                      |
|                                               | Public, 4-year or above       | 440,211          | 133,110                 | 47,093                | 5,134                  | 232,327              | 2,153                    |
| US Service schools                            | Public, 4-year or above       |                  | 201                     | 4                     | 0                      | 27                   |                          |

Cube (in tabular form)

Workspace

# DEMO

What are we going to do?

- Have a first look at Tableau Public
- See how to connect to data sources (and where to find them)
- Have a look at the Tableau Workspace components
- Understand how to create and manage hierarchies
- Navigate OLAP sessions through the various operations to make different analyses